# Resolução - Lista 3

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## Otimização de RNA

Considerando:

$$f(x_1, x_2) = x_1^4 + x_2^4 + x_1^2 x_2 + x_1 x_2^2 - 20x_1^2 - 15x_2^2$$

```
library(plotly)
```

```
## Loading required package: ggplot2
```

```
##
## Attaching package: 'plotly'
```

```
## The following object is masked from 'package:ggplot2':
##
##     last_plot
```

```
## The following object is masked from 'package:stats':
##
##     filter
```

```
## The following object is masked from 'package:graphics':
##
##     layout
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.6
## v forcats   1.0.1     v stringr   1.6.0
## v lubridate 1.9.4     v tibble    3.3.0
## v purrr     1.2.0     v tidyr     1.3.2
```

```
## -- Conflicts ---------------------------------------- tidyverse_conflicts() --
## x dplyr::filter() masks plotly::filter(), stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

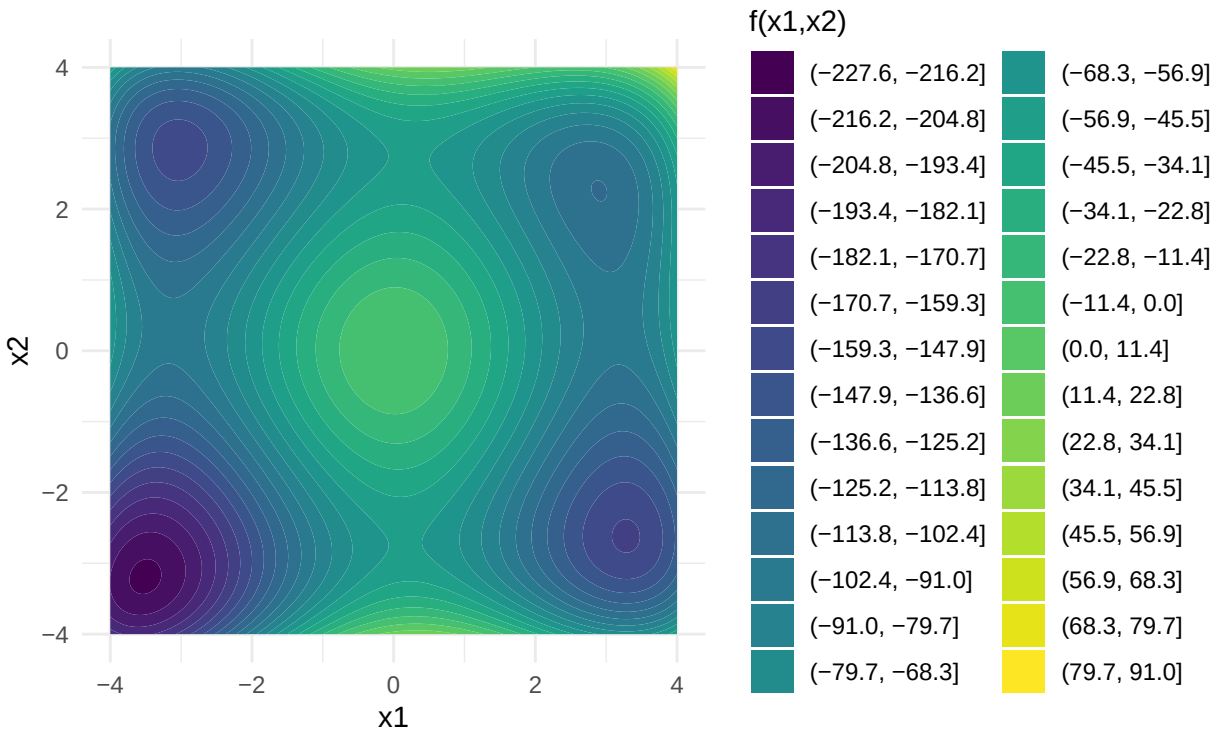
```
f <- function(x1, x2){
  x1^4 + x2^4 + x1^2*x2 + x1*x2^2 - 20*x1^2 - 15*x2^2
}
```

```
# grade (ajuste o intervalo se quiser)
x1 <- seq(-4, 4, length.out = 160)
x2 <- seq(-4, 4, length.out = 160)

# matriz z: linhas = x1, colunas = x2
z <- outer(x1, x2, f)

plot_ly(x = x1, y = x2, z = z, type = "surface") |>
  layout(
    scene = list(
      xaxis = list(title = "x1"),
      yaxis = list(title = "x2"),
      zaxis = list(title = "f(x1,x2)")
    )
  )
```

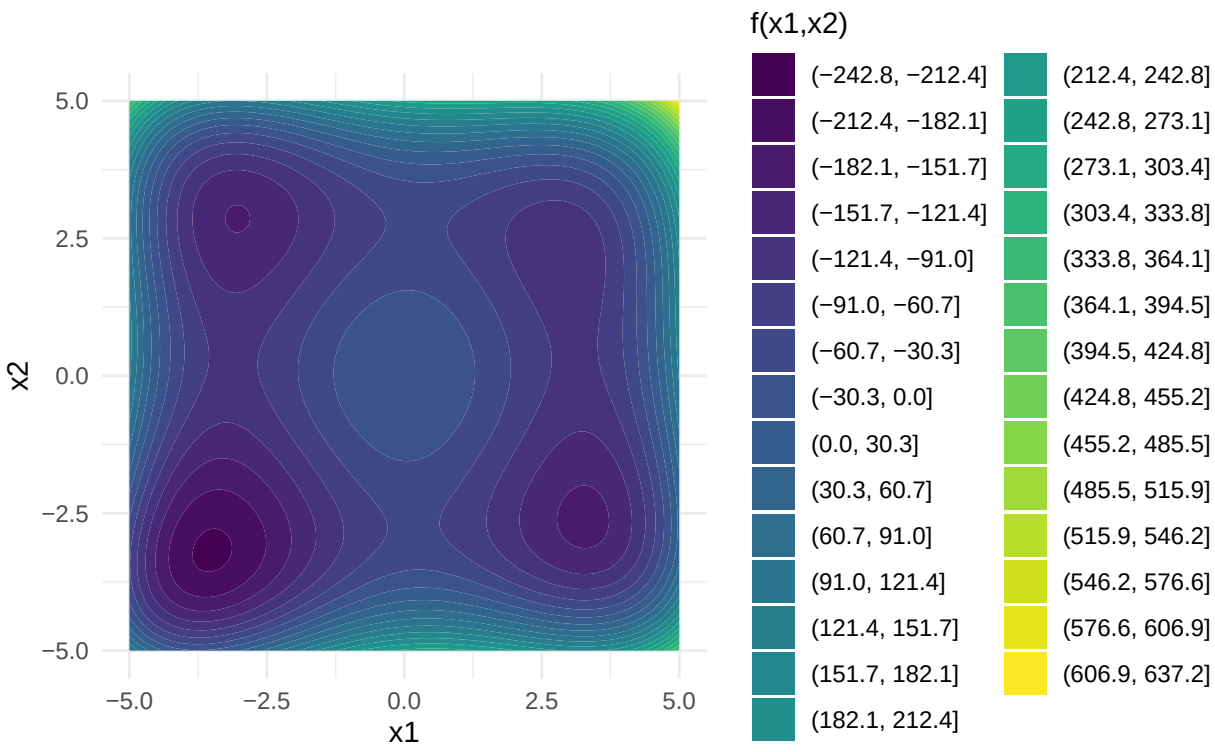
## Letra A

Curvas de nível de f(x1,x2)

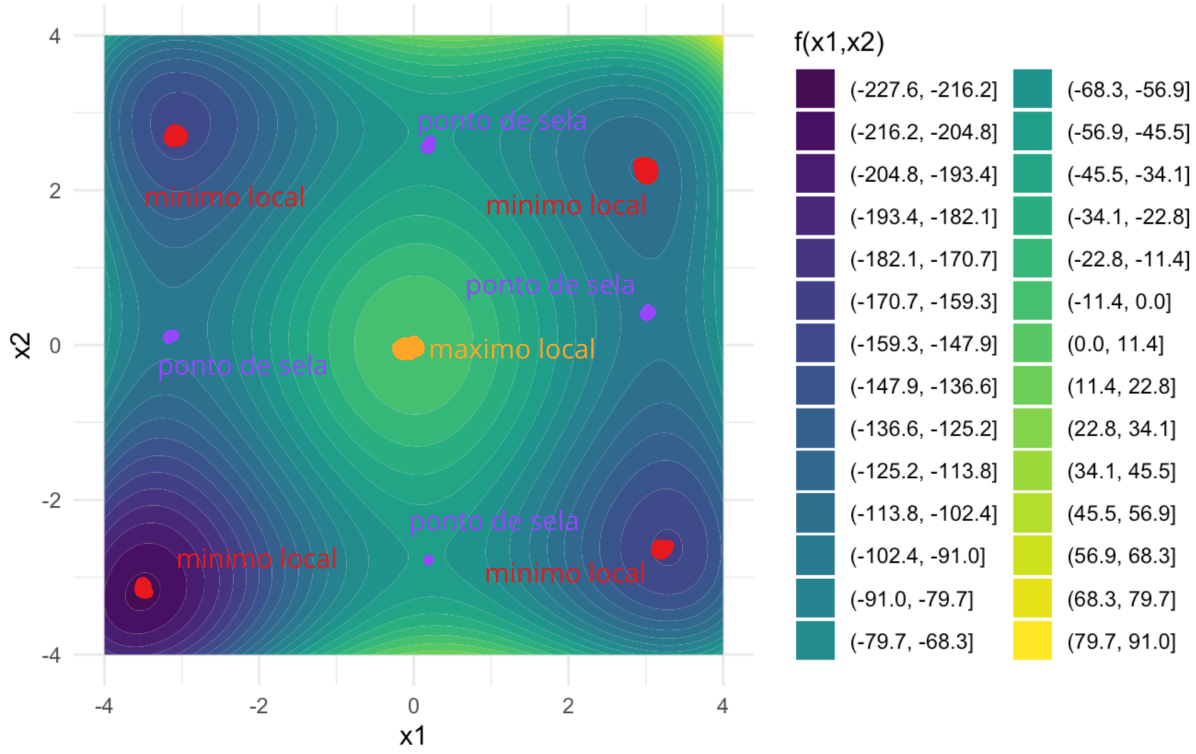
```
grid <- expand.grid(
  x1 = seq(-4, 4, by = 0.02),
  x2 = seq(-4, 4, by = 0.02)
) |>
  as_tibble() |>
  mutate(z = f(x1, x2))

ggplot(grid, aes(x = x1, y = x2, z = z)) +
  geom_contour_filled(bins = 30) +
  coord_equal() +
  labs(x = "x1", y = "x2", fill = "f(x1,x2)") +
  theme_minimal()
```




```
grid <- expand.grid(
  x1 = seq(-5, 5, by = 0.02),
  x2 = seq(-5, 5, by = 0.02)
) |>
  as_tibble() |>
  mutate(z = f(x1, x2))

ggplot(grid, aes(x = x1, y = x2, z = z)) +
  geom_contour_filled(bins = 30) +
  coord_equal() +
  labs(x = "x1", y = "x2", fill = "f(x1,x2)") +
  theme_minimal()
```


f(x1,x2)
(−242.8, −212.4]
(−212.4, −182.1]
(−182.1, −151.7]
(−151.7, −121.4]
(−121.4, −91.0]
(−91.0, −60.7]
(−60.7, −30.3]
(−30.3, 0.0]
(0.0, 30.3]
(30.3, 60.7]
(60.7, 91.0]
(91.0, 121.4]
(121.4, 151.7]
(151.7, 182.1]
(182.1, 212.4]
(212.4, 242.8]
(242.8, 273.1]
(273.1, 303.4]
(303.4, 333.8]
(333.8, 364.1]
(364.1, 394.5]
(394.5, 424.8]
(424.8, 455.2]
(455.2, 485.5]
(485.5, 515.9]
(515.9, 546.2]
(546.2, 576.6]
(576.6, 606.9]
(606.9, 637.2]
x2
5.0
2.5
0.0
−2.5
−5.0
−5.0
−2.5
0.0
2.5
5.0
x1

**Rabiscando na mão os pontos de interesse**




## Letra B

$$\nabla_x f(\boldsymbol{x}) = \left( \frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2} \right).$$

(Ver resolução no papel e caneta)

## Letra C

Função computacional que implementa o método de gradiente para minimizar a função.

Tem que ter:

- taxa de aprendizado
- número de passos

- ponto de partida

Ver resolução prévia que eu fiz na mão para mais detalhes.

```
# derivada parcial 1
df_x1 <- function(x1, x2){
  return(4*x1^3 + 2*x1*x2 + x2^2 - 40*x1)
}

# derivada parcial 2
df_x2 <- function(x1, x2){
  return(4*x2^3 + x1^2 + 2*x1*x2 - 30*x2)
}

# em um k passo, taxa de aprendizado multiplicado por derivadas parciais naquele valor k
grad_step <- function(alpha, x1_k, x2_k){

  g1 <- df_x1(x1_k, x2_k)
  g2 <- df_x2(x1_k, x2_k)

  return(list(
    result_1 = alpha * g1,
    result_2 = alpha * g2
  ))
}
```

```
# testando
print(df_x1(0,5))
```

```
## [1] 25
```

```
print(df_x2(0,5))
```

```
## [1] 350
```

```
a <- grad_step(0.01,0,5)$result_1
print(a)
```

```
## [1] 0.25
```

```
b <- grad_step(0.01,0,5)$result_2
print(b)
```

```
## [1] 3.5
```

```
gradiente_min <- function(alpha, n_passos, x1_init, x2_init){

  hist_temp <- list()

  # inicializo x1 e x2 com os iniciais
```

```
  x1 <- x1_init
  x2 <- x2_init

  hist_temp[[1]] <- c(x1, x2)

  for (i in 1:n_passos) {

    # calculo alpha * derivada parcial 1 e alpha * derivada parcial 2
    step <- grad_step(alpha, x1, x2)

    # atualizo x1 e x2 com essa atualização de valor
    x1 <- x1 - step$result_1
    x2 <- x2 - step$result_2

    # histórico
    hist_temp[[i+1]] <- c(x1, x2)

  }

  hist <- do.call(rbind, hist_temp)
  colnames(hist) <- c("x1", "x2")

  return(list(
    hist_min = hist,
    x1_min = x1,
    x2_min = x2
  ))
}
```

### Letra D

x1_init = 0

x2_init = 5

alpha = 0.01

n_passos = 100

```
result <- gradiente_min(0.01, 100, 0, 5)
```

```
result$x1_min
```

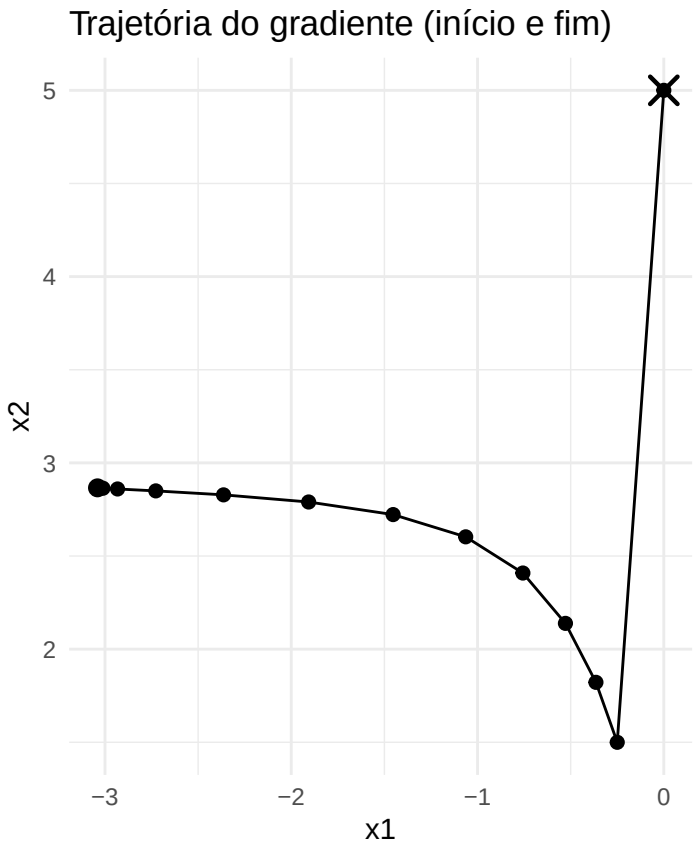
```
## [1] -3.040141
```

```
result$x2_min
```

```
## [1] 2.865981
```

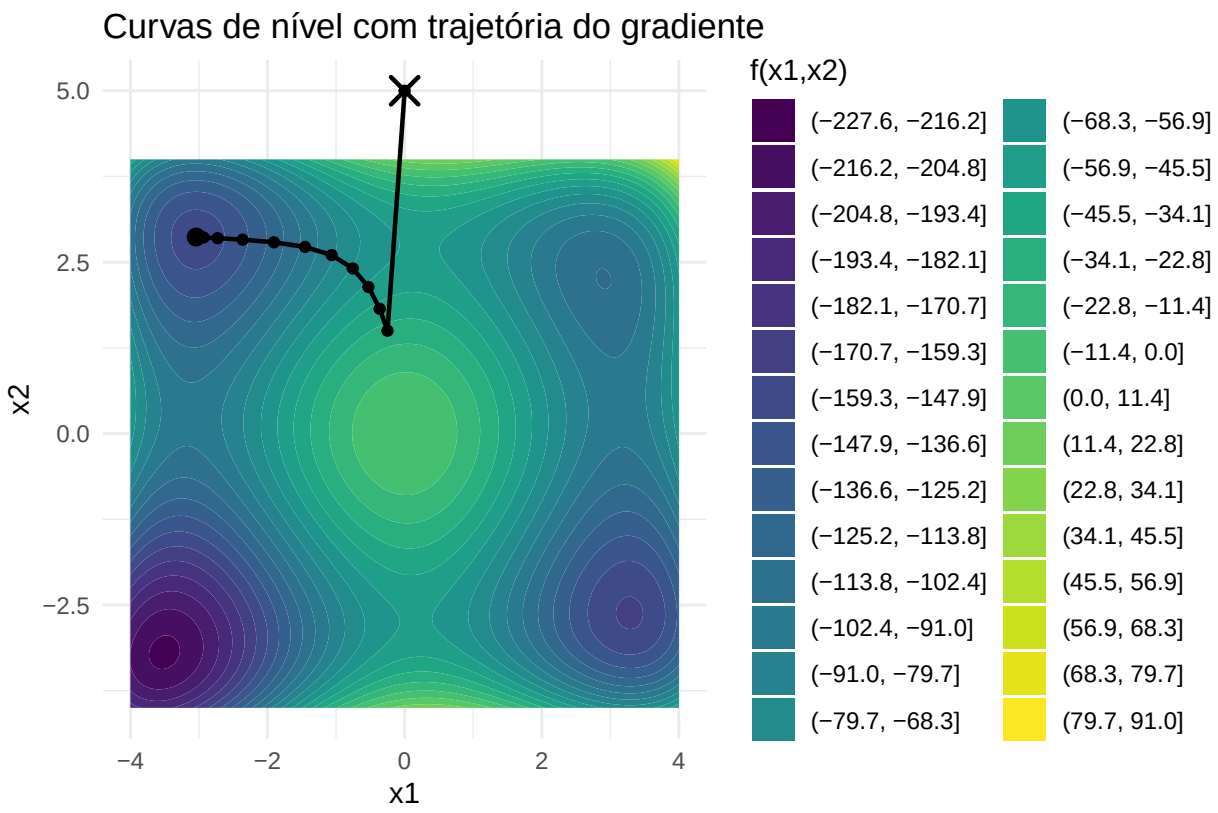
```
hist_df <- as_tibble(result$hist_min) |>
  mutate(iter = row_number())
```

```r
# trajetória (x1 vs x2)
ggplot(hist_df, aes(x = x1, y = x2)) +
  geom_path() +
  geom_point(size = 2) +
  annotate(
    "point",
    x = hist_df$x1[1],
    y = hist_df$x2[1],
    shape = 4,
    size = 4,
    stroke = 1.2
  ) +
  annotate(
    "point",
    x = hist_df$x1[nrow(hist_df)],
    y = hist_df$x2[nrow(hist_df)],
    shape = 16,
    size = 3
  ) +
  coord_equal() +
  labs(
    x = "x1",
    y = "x2",
    title = "Trajetória do gradiente (início e fim)"
  ) +
  theme_minimal()
```




```
# grade das curvas de nível
grid <- expand.grid(
  x1 = seq(-4, 4, by = 0.02),
  x2 = seq(-4, 4, by = 0.02)
) |>
  as_tibble() |>
  mutate(z = f(x1, x2))

ggplot(grid, aes(x = x1, y = x2, z = z)) +
  geom_contour_filled(bins = 30) +

  # trajetória do gradiente simples
  geom_path(
    data = hist_df,
    aes(x = x1, y = x2),
    inherit.aes = FALSE,
    color = "black",
    linewidth = 0.8
  ) +
  geom_point(
    data = hist_df,
    aes(x = x1, y = x2),
    inherit.aes = FALSE,
    color = "black",
    size = 1.5
  ) +
```

```r
# início
annotate(
  "point",
  x = hist_df$x1[1],
  y = hist_df$x2[1],
  shape = 4,
  size = 4,
  stroke = 1.2,
  color = "black"
) +

# fim
annotate(
  "point",
  x = hist_df$x1[nrow(hist_df)],
  y = hist_df$x2[nrow(hist_df)],
  shape = 16,
  size = 3,
  color = "black"
) +

coord_equal() +
labs(
  x = "x1",
  y = "x2",
  fill = "f(x1,x2)",
  title = "Curvas de nível com trajetória do gradiente"
) +
theme_minimal()
```




### Letra E

Agora variando taxa de aprendizado:

- 1
- 0.1
- 0.01
- 0.001
- 0.0001

```
result1 <- gradiente_min(1, 100, 0, 5)
result2 <- gradiente_min(0.1, 100, 0, 5)
result3 <- gradiente_min(0.01, 100, 0, 5)
result4 <- gradiente_min(0.001, 100, 0, 5)
result5 <- gradiente_min(0.0001, 100, 0, 5)
```

```
hist_df1 <- as_tibble(result1$hist_min) |>
  mutate(iter = row_number())

hist_df2 <- as_tibble(result2$hist_min) |>
  mutate(iter = row_number())
```

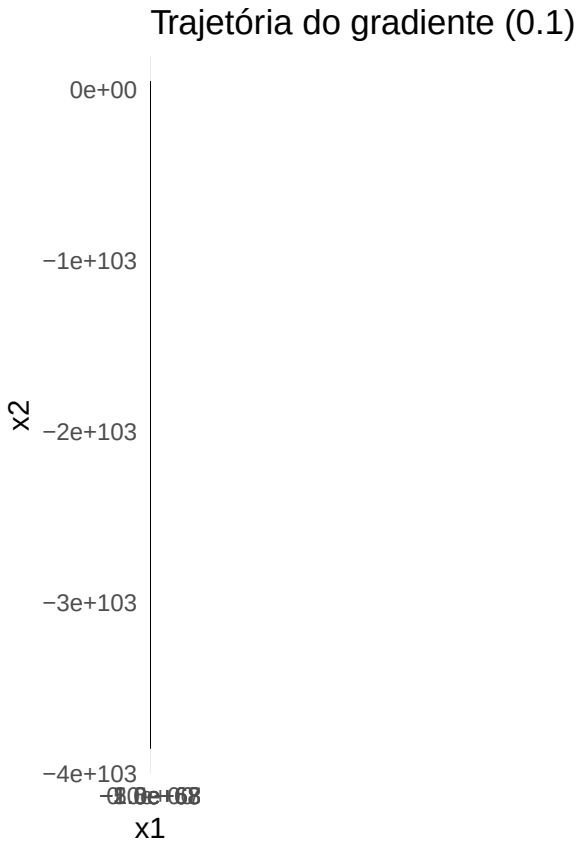
```r
hist_df3 <- as_tibble(result3$hist_min) |>
  mutate(iter = row_number())

hist_df4 <- as_tibble(result4$hist_min) |>
  mutate(iter = row_number())

hist_df5 <- as_tibble(result5$hist_min) |>
  mutate(iter = row_number())

# 1


hist_df1 <- hist_df1 |>
  filter(is.finite(x1), is.finite(x2))

ggplot(hist_df1, aes(x = x1, y = x2)) +
  geom_path() +
  geom_point(size = 2) +
  annotate(
    "point",
    x = hist_df1$x1[1],
    y = hist_df1$x2[1],
    shape = 4,
    size = 4,
    stroke = 1.2
  ) +
  annotate(
    "point",
    x = hist_df1$x1[nrow(hist_df)],
    y = hist_df1$x2[nrow(hist_df)],
    shape = 16,
    size = 3
  ) +
  coord_equal() +
  labs(
    x = "x1",
    y = "x2",
    title = "Trajetória do gradiente (1)"
  ) +
  theme_minimal()
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```


```
# 0.1

hist_df2 <- hist_df2 |>
  filter(is.finite(x1), is.finite(x2))

ggplot(hist_df2, aes(x = x1, y = x2)) +
  geom_path() +
  geom_point(size = 2) +
  annotate(
    "point",
    x = hist_df2$x1[1],
    y = hist_df2$x2[1],
    shape = 4,
    size = 4,
    stroke = 1.2
  ) +
  annotate(
    "point",
    x = hist_df2$x1[nrow(hist_df)],
    y = hist_df2$x2[nrow(hist_df)],
    shape = 16,
    size = 3
  ) +
  coord_equal() +
  labs(
    x = "x1",
```

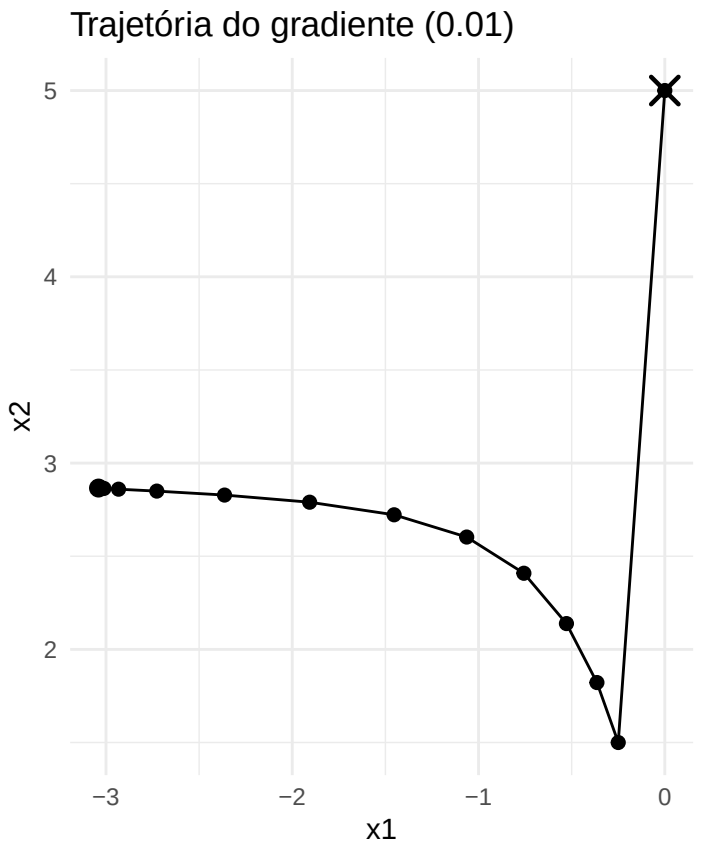
```
    y = "x2",
    title = "Trajetória do gradiente (0.1)"
  ) +
  theme_minimal()
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```



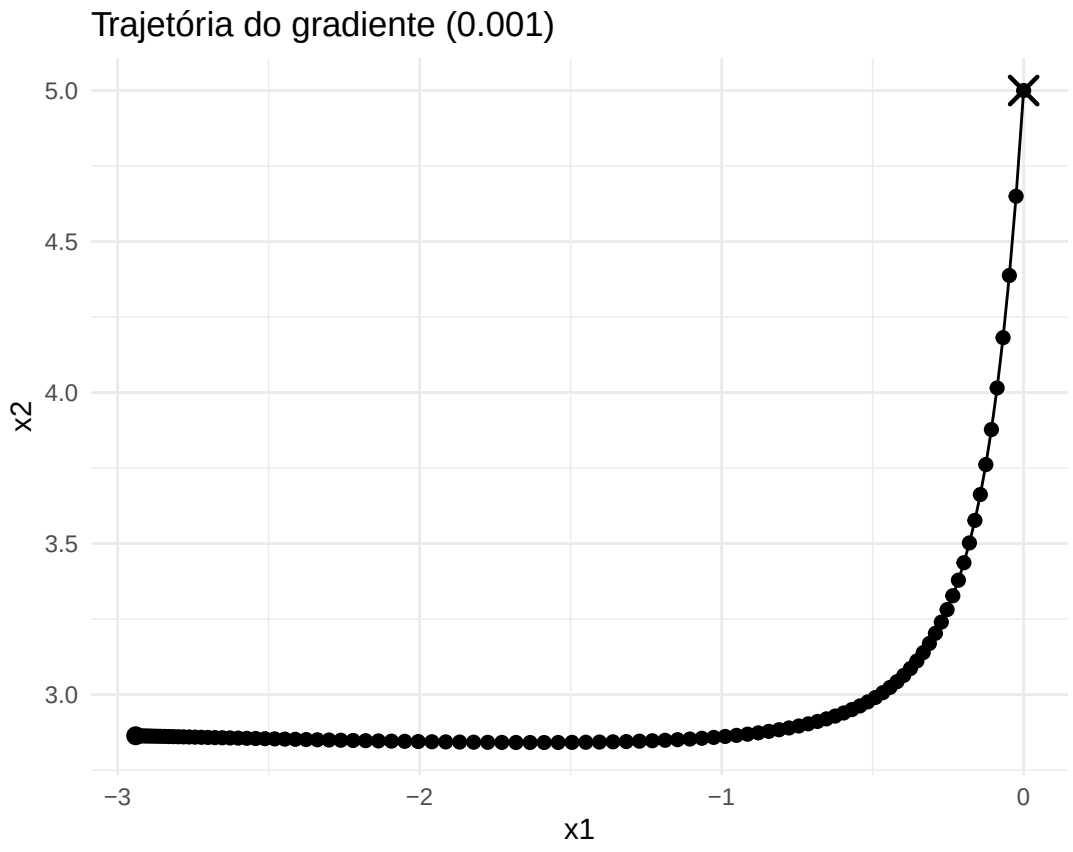

```
# 0.01
ggplot(hist_df3, aes(x = x1, y = x2)) +
  geom_path() +
  geom_point(size = 2) +
  annotate(
    "point",
    x = hist_df3$x1[1],
    y = hist_df3$x2[1],
    shape = 4,
    size = 4,
    stroke = 1.2
  ) +
  annotate(
    "point",
    x = hist_df3$x1[nrow(hist_df)],
    y = hist_df3$x2[nrow(hist_df)],
    shape = 16,
```

```
    size = 3
  ) +
  coord_equal() +
  labs(
    x = "x1",
    y = "x2",
    title = "Trajetória do gradiente (0.01)"
  ) +
  theme_minimal()
```




```
# 0.001

hist_df4 <- hist_df4 |>
  filter(is.finite(x1), is.finite(x2))

ggplot(hist_df4, aes(x = x1, y = x2)) +
  geom_path() +
  geom_point(size = 2) +
  annotate(
    "point",
    x = hist_df4$x1[1],
    y = hist_df4$x2[1],
    shape = 4,
    size = 4,
    stroke = 1.2
```

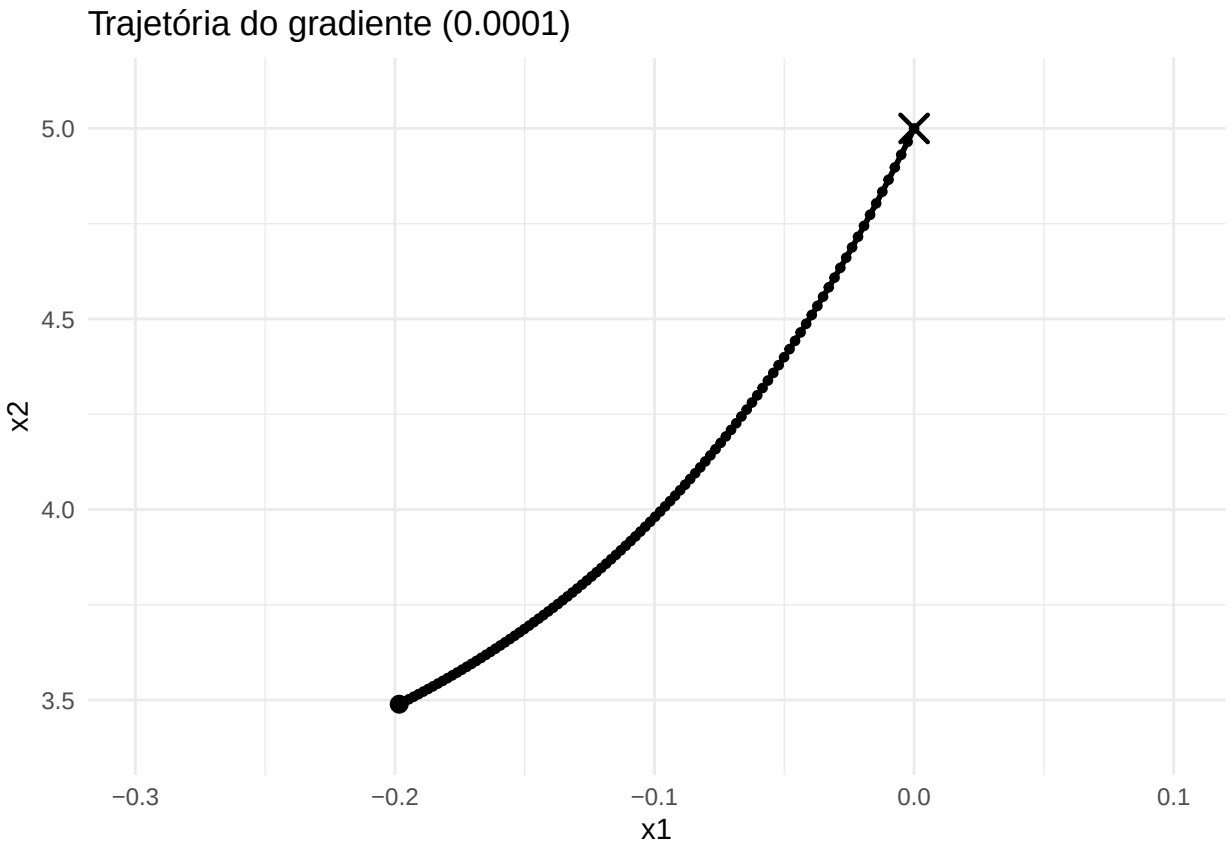
```
  ) +
  annotate(
    "point",
    x = hist_df4$x1[nrow(hist_df)],
    y = hist_df4$x2[nrow(hist_df)],
    shape = 16,
    size = 3
  ) +
  coord_equal() +
  labs(
    x = "x1",
    y = "x2",
    title = "Trajetória do gradiente (0.001)"
  ) +
  theme_minimal()
```




```
# 0.0001

# limpeza mínima (evita NA/Inf)
hist_df5_ok <- hist_df5 |>
  filter(is.finite(x1), is.finite(x2))

n <- nrow(hist_df5_ok)

ggplot(hist_df5_ok, aes(x = x1, y = x2)) +
```

```
  geom_path(linewidth = 1) +
  geom_point(size = 1.2) +
  annotate(
    "point",
    x = hist_df5_ok$x1[1],
    y = hist_df5_ok$x2[1],
    shape = 4,
    size = 4,
    stroke = 1.2
  ) +
  annotate(
    "point",
    x = hist_df5_ok$x1[n],
    y = hist_df5_ok$x2[n],
    shape = 16,
    size = 3
  ) +
  coord_equal() +
  coord_cartesian(
    xlim = range(hist_df5_ok$x1) + c(-0.1, 0.1),
    ylim = range(hist_df5_ok$x2) + c(-0.1, 0.1)
  ) +
  labs(
    x = "x1",
    y = "x2",
    title = "Trajetória do gradiente (0.0001)"
  ) +
  theme_minimal()
```

```
## Coordinate system already present.
## i Adding new coordinate system, which will replace the existing one.
```




Taxas de aprendizado

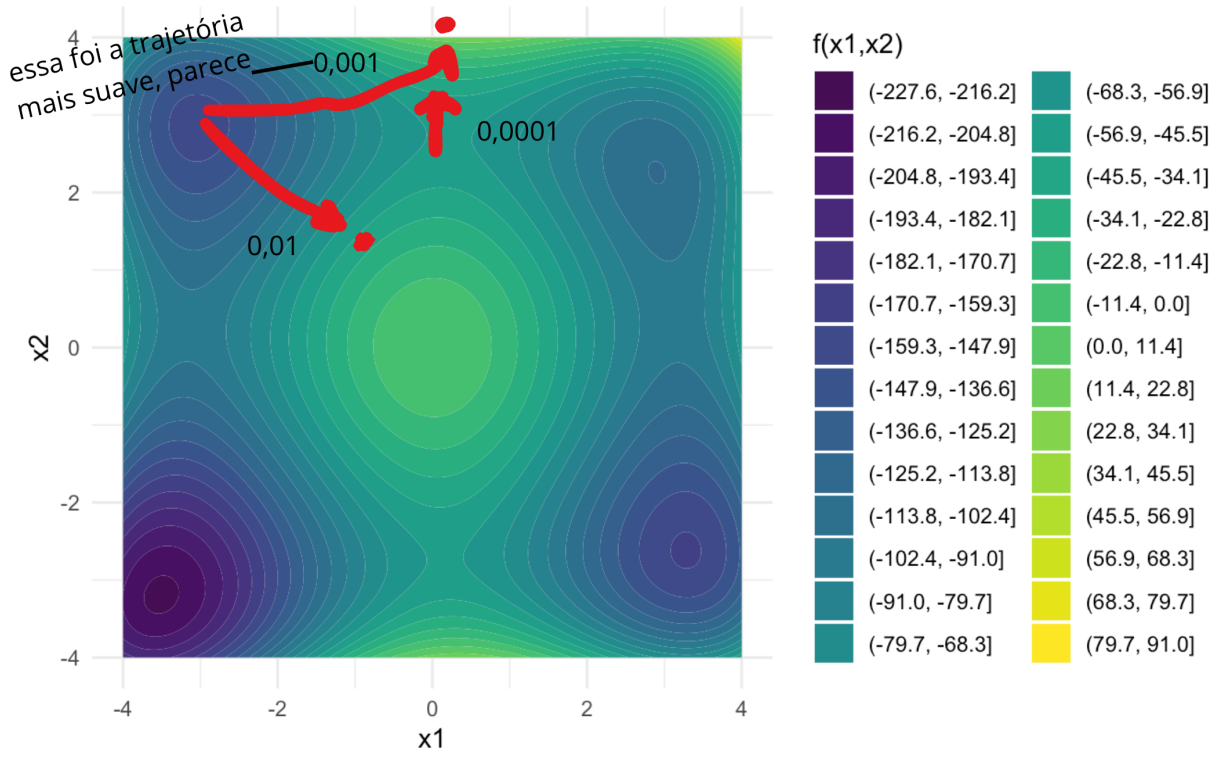
1 - divergiu

0.1 - divergiu

0.01 - passo muito grande?

0.001 - mais sauve, segue curva do vale <- esse seria o valor escolhido!

0.0001 - mais lento, estável




### Letra F

```r
set.seed(123)

iniciais <- tibble(
  id = 1:20,
  x1_init = runif(20, min = -5, max = 5),
  x2_init = runif(20, min = -5, max = 5)
)
```

```r
alpha <- 0.001
n_passos <- 5000

# objetos para guardar resultados finais
x1_final <- numeric(20)
x2_final <- numeric(20)

hist_all <- data.frame()

for (i in 1:20) {

  res <- gradiente_min(
    alpha = alpha,
    n_passos = n_passos,
    x1_init = iniciais$x1_init[i],
    x2_init = iniciais$x2_init[i]
  )
```

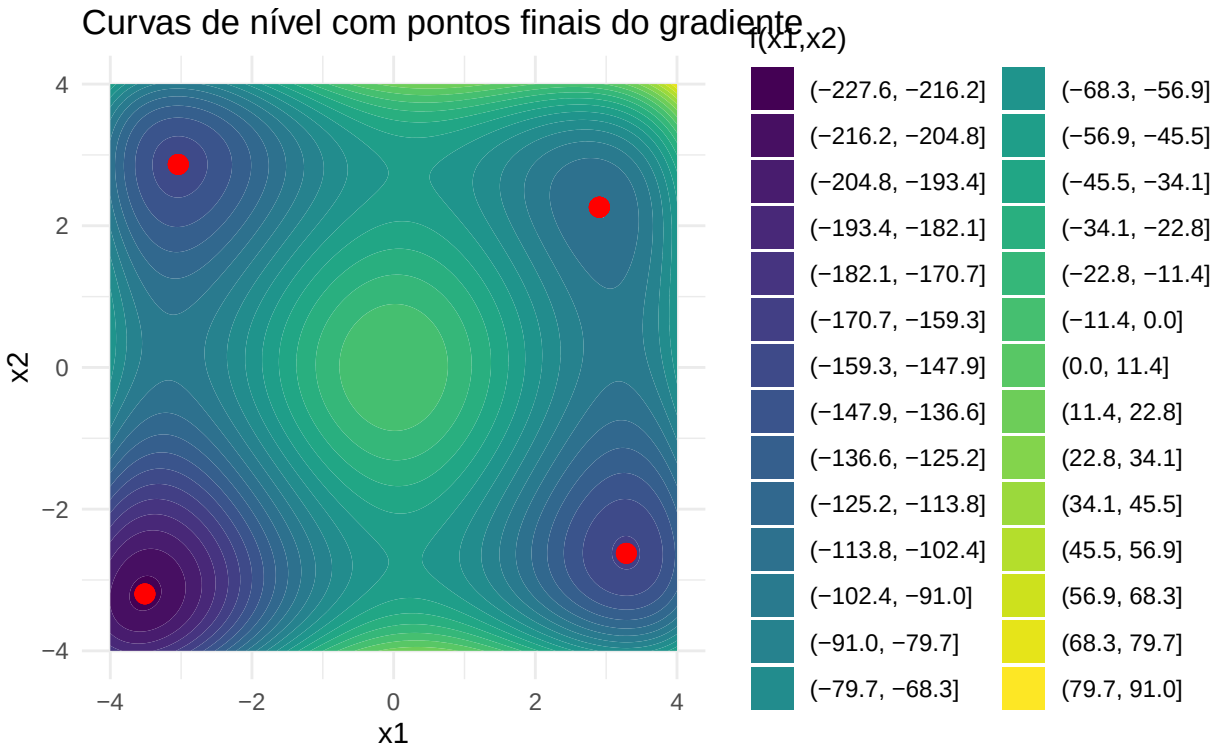
```
  x1_final[i] <- res$x1_min
  x2_final[i] <- res$x2_min

  hist_i <- as.data.frame(res$hist_min)
  colnames(hist_i) <- c("x1", "x2")

  hist_i$run  <- i
  hist_i$iter <- 1:nrow(hist_i)

  hist_all <- rbind(hist_all, hist_i)
}
```

```
resultados <- data.frame(
  x1_init = iniciais$x1_init,
  x2_init = iniciais$x2_init,
  x1_final = x1_final,
  x2_final = x2_final
)

resultados
```

```
##       x1_init    x2_init  x1_final  x2_final
## 1  -2.1242248  3.8953932 -3.040141  2.865981
## 2   2.8830514  1.9280341  2.903159  2.262064
## 3  -0.9102308  1.4050681 -3.040141  2.865981
## 4   3.8301740  4.9426978  2.903159  2.262064
## 5   4.4046728  1.5570580  2.903159  2.262064
## 6  -4.5444350  2.0853047 -3.040141  2.865981
## 7   0.2810549  0.4406602  2.903159  2.262064
## 8   3.9241904  0.9414202  2.903159  2.262064
## 9   0.5143501 -2.1084026  3.284497 -2.624033
## 10 -0.4338526 -3.5288635 -3.510853 -3.196775
## 11  4.5683335  4.6302423  2.903159  2.262064
## 12 -0.4666584  4.0229905 -3.040141  2.865981
## 13  1.7757064  1.9070528  2.903159  2.262064
## 14  0.7263340  2.9546742  2.903159  2.262064
## 15 -3.9707532 -4.7538632 -3.510853 -3.196775
## 16  3.9982497 -0.2220403  3.284497 -2.624033
## 17 -2.5391227  2.5845954 -3.040141  2.865981
## 18 -4.5794047 -2.8359206 -3.510853 -3.196775
## 19 -1.7207928 -1.8181899 -3.510853 -3.196775
## 20  4.5450365 -2.6837421  3.284497 -2.624033
```

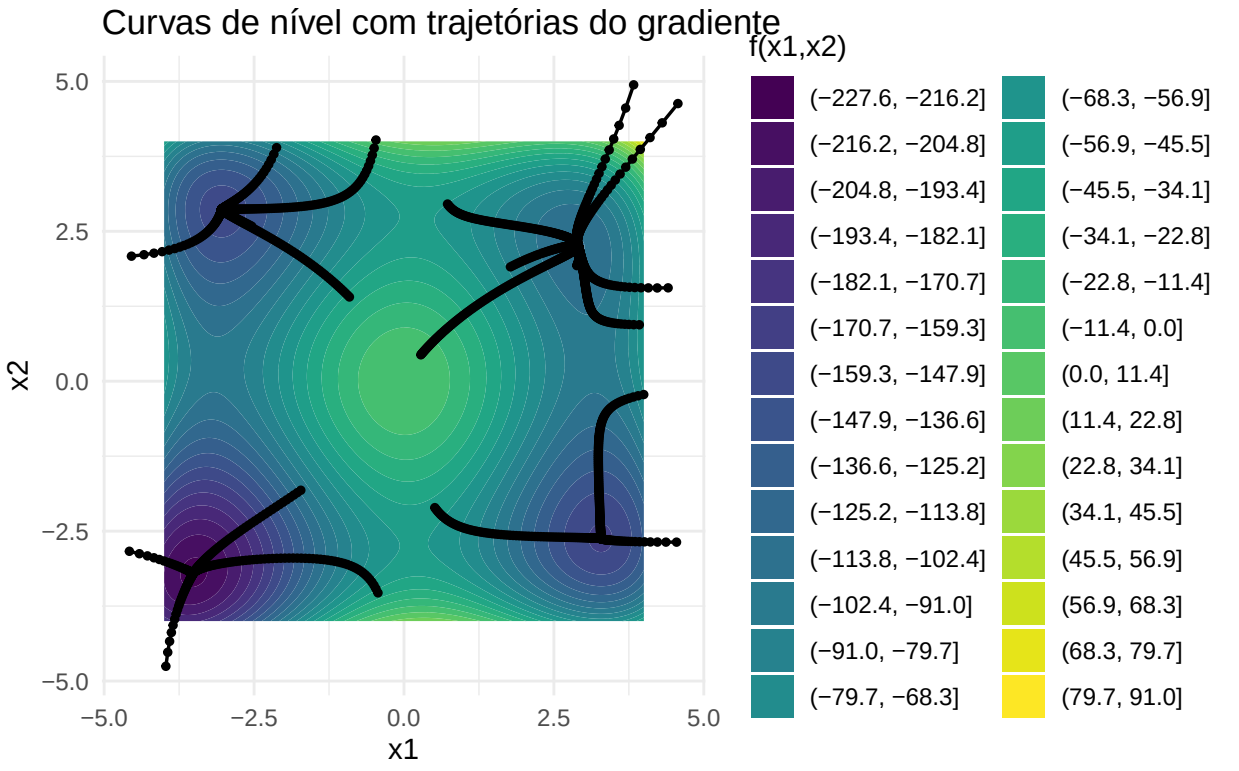
```
ggplot(grid, aes(x = x1, y = x2, z = z)) +
  geom_contour_filled(bins = 30) +
  geom_point(
    data = resultados,
    aes(x = x1_final, y = x2_final),
    inherit.aes = FALSE,
    color = "red",
    size = 3
  ) +
```

```
  coord_equal() +
  labs(
    x = "x1",
    y = "x2",
    fill = "f(x1,x2)",
    title = "Curvas de nível com pontos finais do gradiente"
  ) +
  theme_minimal()
```




```
ggplot(grid, aes(x = x1, y = x2, z = z)) +
  geom_contour_filled(bins = 30) +
  geom_path(
    data = hist_all,
    aes(x = x1, y = x2, group = run),
    inherit.aes = FALSE,
    color = "black",
    linewidth = 0.6
  ) +
  geom_point(
    data = hist_all,
    aes(x = x1, y = x2, group = run),
    inherit.aes = FALSE,
    color = "black",
    size = 1
  ) +
```

```
  coord_equal() +
  labs(
    x = "x1",
    y = "x2",
    fill = "f(x1,x2)",
    title = "Curvas de nível com trajetórias do gradiente"
  ) +
  theme_minimal()
```




Todos convergem, mas nem todos convergem para o mesmo mínimo!

## Letra G

Repetindo letra D, fazendo gradiente com momento.

O método do gradiente com momento incorpora uma média exponencial dos gradientes passados, reduzindo oscilações e acelerando a convergência em direções de curvatura suave.

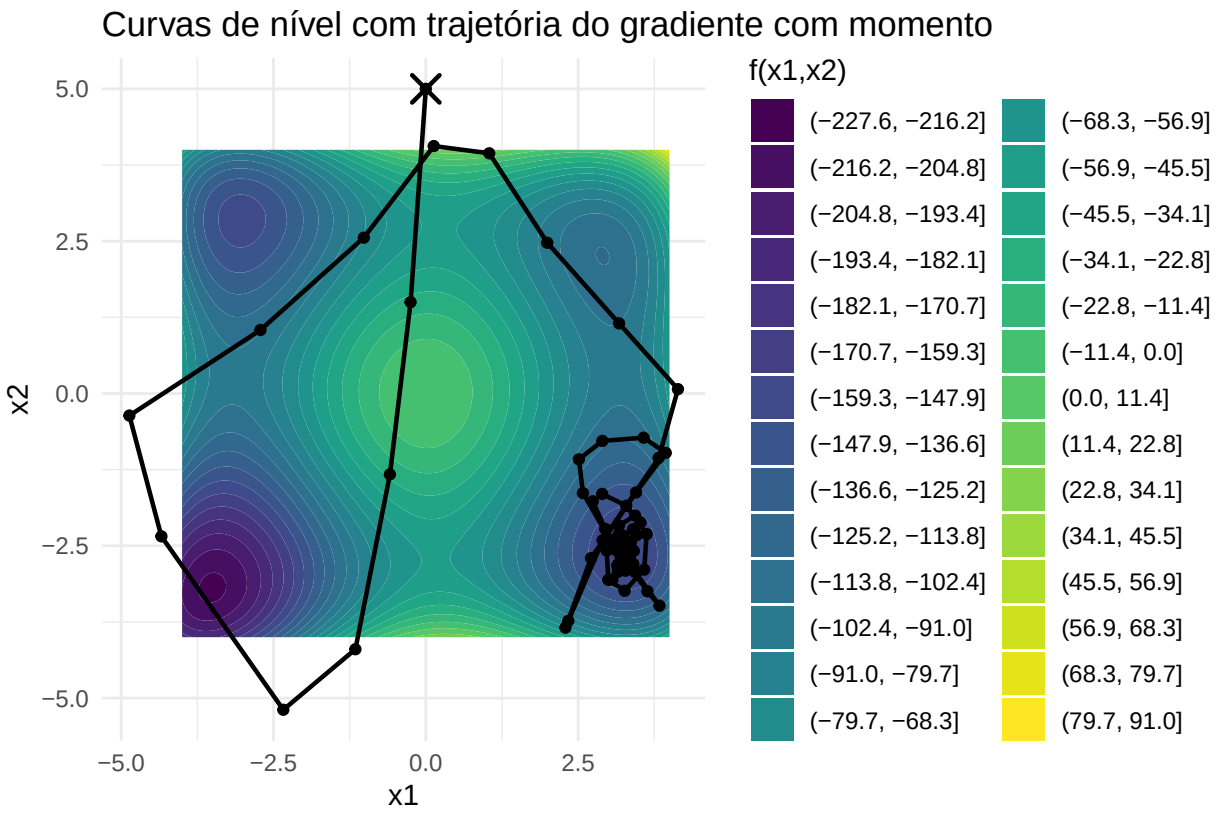
```
# paramentro momento entre 0 e 1
gradiente_momento <- function(alpha, momentum, n_passos, x1_init, x2_init){

  hist_temp <- list()

  # inicializo x1 e x2 com os iniciais
  x1 <- x1_init
```

```
  x2 <- x2_init

  v1 <- 0
  v2 <- 0

  hist_temp[[1]] <- c(x1, x2)

  for (i in 1:n_passos) {

    # calculo alpha * derivada parcial 1 e alpha * derivada parcial 2
    step <- grad_step(alpha, x1, x2)


    v1 <- (momentum * v1) + df_x1(x1, x2)
    v2 <- (momentum * v2) + df_x2(x1, x2)

    # atualizo x1 e x2 com essa atualização de valor
    x1 <- x1 - alpha * v1
    x2 <- x2 - alpha * v2

    # histórico
    hist_temp[[i+1]] <- c(x1, x2)

  }

  hist <- do.call(rbind, hist_temp)
  colnames(hist) <- c("x1", "x2")

  return(list(
    hist_min = hist,
    x1_min = x1,
    x2_min = x2
  ))
}
```

```
result_momento <- gradiente_momento(0.01, 0.9, 100, 0, 5)
```

```
hist_df <- as_tibble(result_momento$hist_min) |>
  mutate(iter = row_number())

# grade das curvas de nível
grid <- expand.grid(
  x1 = seq(-4, 4, by = 0.02),
  x2 = seq(-4, 4, by = 0.02)
) |>
  as_tibble() |>
  mutate(z = f(x1, x2))

ggplot(grid, aes(x = x1, y = x2, z = z)) +
  geom_contour_filled(bins = 30) +

  # trajetória do gradiente com momento
  geom_path(
```

```r
  data = hist_df,
  aes(x = x1, y = x2),
  inherit.aes = FALSE,
  color = "black",
  linewidth = 0.8
) +
geom_point(
  data = hist_df,
  aes(x = x1, y = x2),
  inherit.aes = FALSE,
  color = "black",
  size = 1.5
) +

# início (X)
annotate(
  "point",
  x = hist_df$x1[1],
  y = hist_df$x2[1],
  shape = 4,
  size = 4,
  stroke = 1.2,
  color = "black"
) +

annotate(
  "point",
  x = hist_df$x1[nrow(hist_df)],
  y = hist_df$x2[nrow(hist_df)],
  shape = 16,
  size = 3,
  color = "black"
) +

coord_equal() +
labs(
  x = "x1",
  y = "x2",
  fill = "f(x1,x2)",
  title = "Curvas de nível com trajetória do gradiente com momento"
) +
theme_minimal()
```




Eita, o gradiente com momento foi para OUTRO mínimo local que não o do canto superior esquerdo.

### Letra H

Repetindo letra D, fazendo gradiente RMSProp

```
# taxa de aprendizado "alpha" = 0.001
# decaimento "r" entre 0 e 1 = 0.9
# pequeno "e" constante = 0.0000001
gradiente_rms_prop <- function(alpha, r, e, x1_init, x2_init){

  hist_temp <- list()

  qtd_exec <- 1

  # inicializo x1 e x2 com os iniciais
  x1 <- x1_init
  x2 <- x2_init

  hist_temp[[qtd_exec]] <- c(x1, x2)

  # media exponencial
  s1 <- 0
  s2 <- 0
```

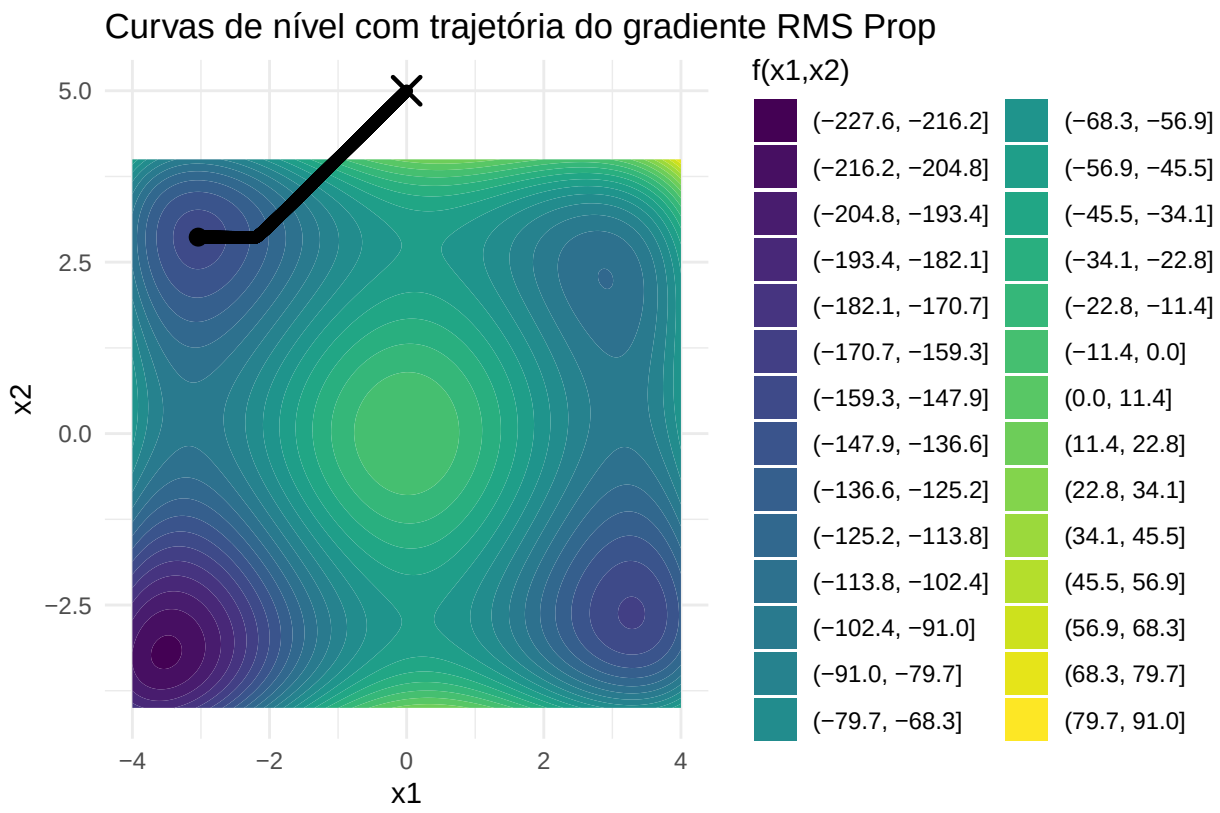
```
  # o loop segue enquanto n bate condição de parada
  i <- TRUE

  while (i == TRUE) {

    # calculo alpha * derivada parcial 1 e alpha * derivada parcial 2
    step <- grad_step(alpha, x1, x2)

    # atualiza a media exponencial
    s1 <- (r * s1) + ((1 - r) * (step$result_1 * step$result_1))
    s2 <- (r * s2) + ((1 - r) * (step$result_2 * step$result_2))

    # atualizo x1 e x2 dado os s1 e s2
    x1 <- x1 - (alpha * (step$result_1 / ( sqrt(s1) + e)))

    x2 <- x2 - (alpha * (step$result_2 / ( sqrt(s2) + e)))

    qtd_exec <- qtd_exec + 1

    # histórico
    hist_temp[[qtd_exec]] <- c(x1, x2)

    # vou parar quando gradiente estiver perto de zero
    if ( (sqrt(sum(step$result_1^2)) < e )|| (sqrt(sum(step$result_2^2)) < e)){
      i <- FALSE
    }

  }

  print(qtd_exec)

  hist <- do.call(rbind, hist_temp)
  colnames(hist) <- c("x1", "x2")

  return(list(
    hist_min = hist,
    x1_min = x1,
    x2_min = x2
  ))
}
```

```
result_rms_prop <- gradiente_rms_prop(0.001, 0.9, 0.0000001, 0, 5)
```

```
## [1] 3094
```

```
hist_df <- as_tibble(result_rms_prop$hist_min) |>
  mutate(iter = row_number())

# grade das curvas de nível
grid <- expand.grid(
  x1 = seq(-4, 4, by = 0.02),
  x2 = seq(-4, 4, by = 0.02)
) |>
```

```r
  as_tibble() |>
  mutate(z = f(x1, x2))

ggplot(grid, aes(x = x1, y = x2, z = z)) +
  geom_contour_filled(bins = 30) +

  # trajetória do gradiente com momento
  geom_path(
    data = hist_df,
    aes(x = x1, y = x2),
    inherit.aes = FALSE,
    color = "black",
    linewidth = 0.8
  ) +
  geom_point(
    data = hist_df,
    aes(x = x1, y = x2),
    inherit.aes = FALSE,
    color = "black",
    size = 1.5
  ) +

  # início (X)
  annotate(
    "point",
    x = hist_df$x1[1],
    y = hist_df$x2[1],
    shape = 4,
    size = 4,
    stroke = 1.2,
    color = "black"
  ) +

  # fim ( )
  annotate(
    "point",
    x = hist_df$x1[nrow(hist_df)],
    y = hist_df$x2[nrow(hist_df)],
    shape = 16,
    size = 3,
    color = "black"
  ) +

  coord_equal() +
  labs(
    x = "x1",
    y = "x2",
    fill = "f(x1,x2)",
    title = "Curvas de nível com trajetória do gradiente RMS Prop"
  ) +
  theme_minimal()
```




Eita que o RMS Prop foi SECO no minimo local.

## Letra I

Repetindo letra D, fazendo gradiente Adam

```
# taxa de aprendizado "alpha" = 0.001
# decaimento "r1" e "r2" entre 0 e 1 -> nesse exemplo 0.9 e 0.99
# pequeno "e" constante = 0.0000001
gradiente_adam <- function(alpha, r1, r2, e, x1_init, x2_init){

  hist_temp <- list()

  qtd_exec <- 1

  # inicializo x1 e x2 com os iniciais
  x1 <- x1_init
  x2 <- x2_init

  # primeiro momento
  m1 <- 0
  m2 <- 0

  # segundo momento
  v1 <- 0
```

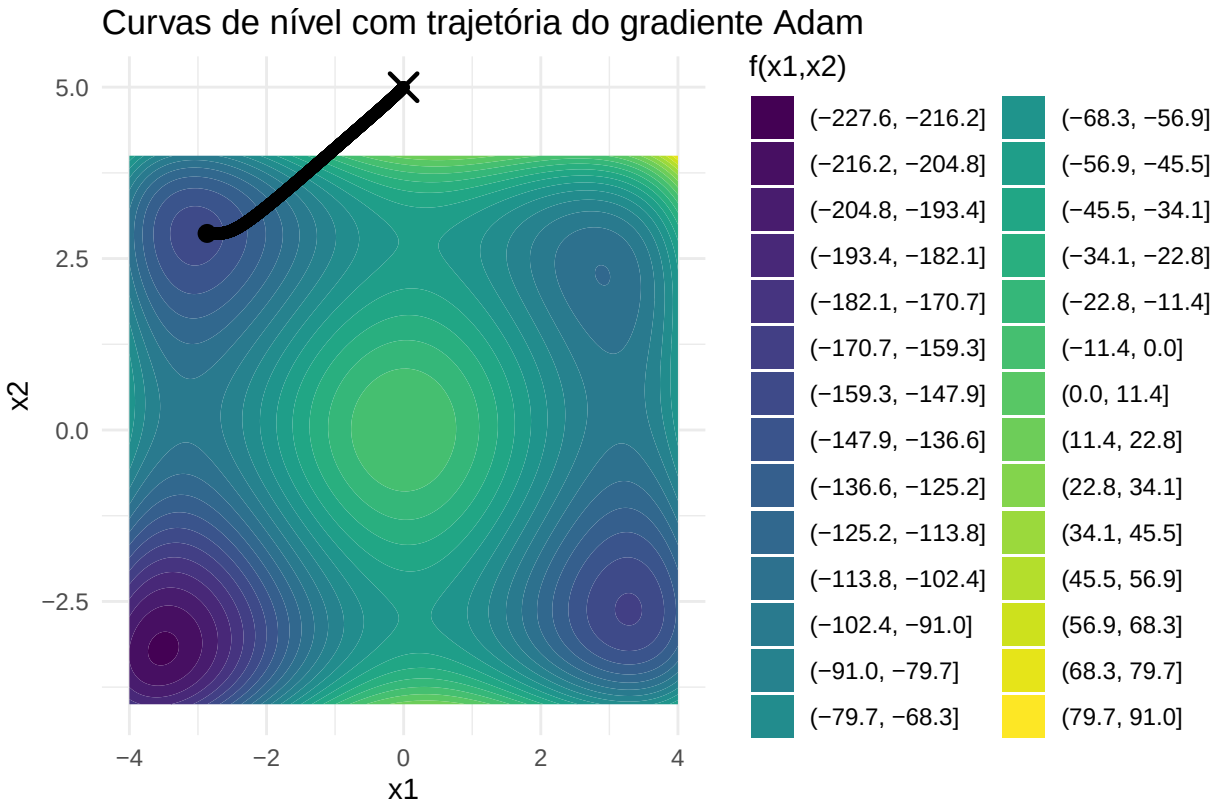
```
  v2 <- 0

  # o loop segue enquanto n bate condição de parada
  i = TRUE

  while (i == TRUE) {

    # calculo alpha * derivada parcial 1 e alpha * derivada parcial 2
    step <- grad_step(alpha, x1, x2)

    # atualiza o primeiro momento
    m1 <- (r1 * m1) + ((1 - r1) * step$result_1)
    m2 <- (r1 * m2) + ((1 - r1) * step$result_2)

    # atualiza o segundo momento
    v1 <- (r2 * v1) + ((1 - r2) * (step$result_1 * step$result_1))
    v2 <- (r2 * v2) + ((1 - r2) * (step$result_2 * step$result_2))

    # correção de viés
    #m1 <- m1/1-r1
    #m2 <- m2/1-r1

    #v1 <- v1/1-r2
    #v2 <- v2/1-r2

    # atualizo x1 e x2 dado os s1 e s2
    x1 <- x1 - (alpha * (m1/(sqrt(v1)+e)))
    x2 <- x2 - (alpha * (m2/(sqrt(v2)+e)))

    qtd_exec <- qtd_exec + 1

    # histórico
    hist_temp[[qtd_exec]] <- c(x1, x2)

    # vou parar quando gradiente estiver perto de zero
    if ( (sqrt(sum(step$result_1^2)) < e )|| (sqrt(sum(step$result_2^2)) < e)){
      i = FALSE
    }

  }

  print(qtd_exec)

  hist <- do.call(rbind, hist_temp)
  colnames(hist) <- c("x1", "x2")

  return(list(
    hist_min = hist,
    x1_min = x1,
    x2_min = x2
  ))
}
```

```r
result_adam <- gradiente_adam(0.001, 0.9, 0.99, 0.0000001, 0, 5)
```

```
## [1] 2832
```

```r
hist_df <- as_tibble(result_adam$hist_min) |>
  mutate(iter = row_number())

# grade das curvas de nível
grid <- expand.grid(
  x1 = seq(-4, 4, by = 0.02),
  x2 = seq(-4, 4, by = 0.02)
) |>
  as_tibble() |>
  mutate(z = f(x1, x2))

ggplot(grid, aes(x = x1, y = x2, z = z)) +
  geom_contour_filled(bins = 30) +

  # trajetória do gradiente com momento
  geom_path(
    data = hist_df,
    aes(x = x1, y = x2),
    inherit.aes = FALSE,
    color = "black",
    linewidth = 0.8
  ) +
  geom_point(
    data = hist_df,
    aes(x = x1, y = x2),
    inherit.aes = FALSE,
    color = "black",
    size = 1.5
  ) +

  # início (X)
  annotate(
    "point",
    x = hist_df$x1[1],
    y = hist_df$x2[1],
    shape = 4,
    size = 4,
    stroke = 1.2,
    color = "black"
  ) +

  # fim ( )
  annotate(
    "point",
    x = hist_df$x1[nrow(hist_df)],
    y = hist_df$x2[nrow(hist_df)],
    shape = 16,
    size = 3,
    color = "black"
```

```
  ) +

  coord_equal() +
  labs(
    x = "x1",
    y = "x2",
    fill = "f(x1,x2)",
    title = "Curvas de nível com trajetória do gradiente Adam"
  ) +
  theme_minimal()
```




Resumindo, Adam e RMS Prop foram mais interessantes. Foram seco no ponto.

### Letra J

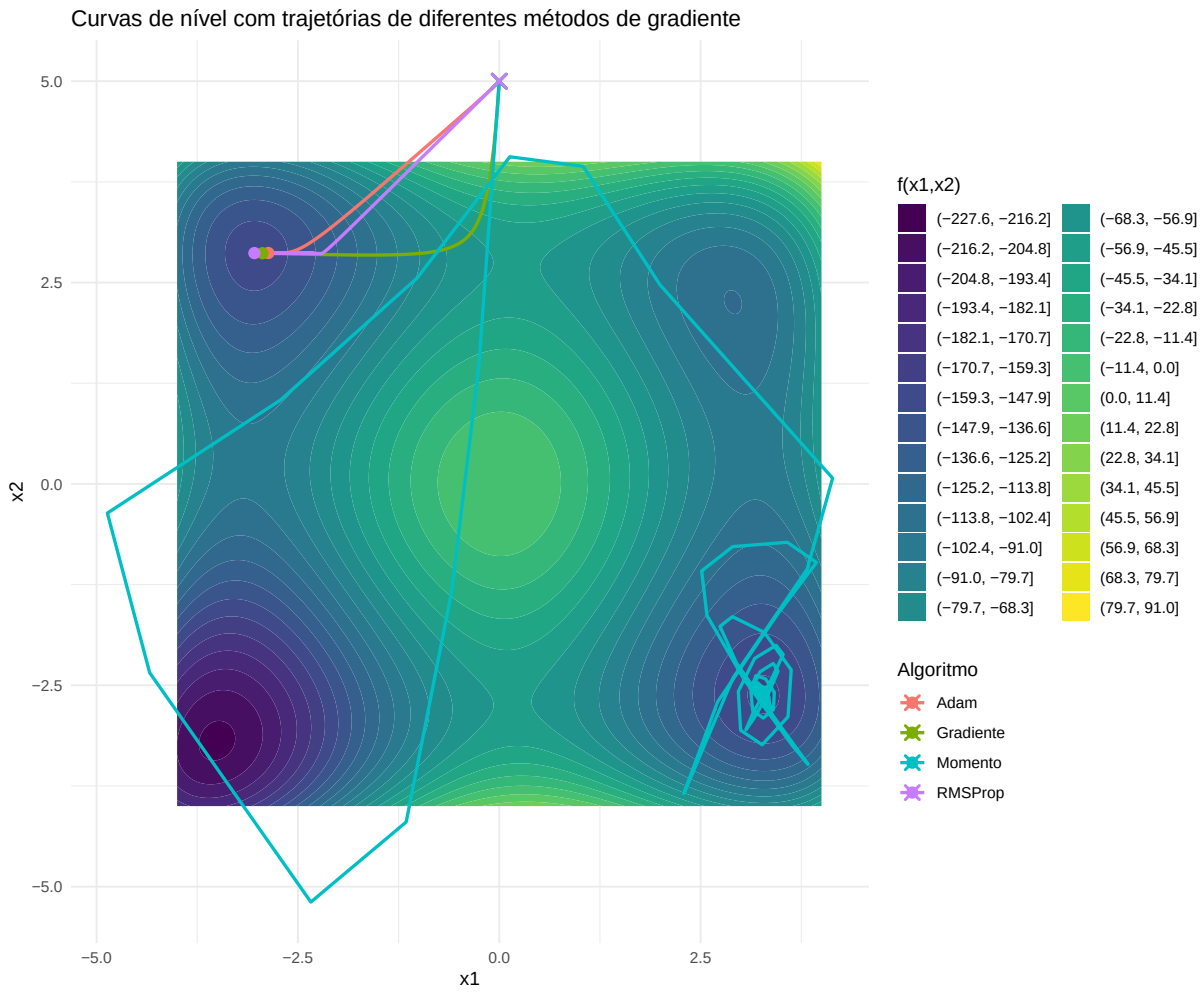
Grafico dos passos nas letras d, g, h, i tudo junto.

```
hist_all <- bind_rows(
  as_tibble(result_adam$hist_min) |>
    mutate(algoritmo = "Adam"),

  as_tibble(result_rms_prop$hist_min) |>
    mutate(algoritmo = "RMSProp"),

  as_tibble(result_momento$hist_min) |>
```

```
    mutate(algoritmo = "Momento"),

  as_tibble(result4$hist_min) |>
    mutate(algoritmo = "Gradiente")
) |>
  group_by(algoritmo) |>
  mutate(iter = row_number()) |>
  ungroup() |>
  filter(is.finite(x1), is.finite(x2))
```

```
grid <- expand.grid(
  x1 = seq(-4, 4, by = 0.02),
  x2 = seq(-4, 4, by = 0.02)
) |>
  as_tibble() |>
  mutate(z = f(x1, x2))
```

```
ggplot(grid, aes(x = x1, y = x2, z = z)) +
  geom_contour_filled(bins = 30) +

  geom_path(
    data = hist_all,
    aes(x = x1, y = x2, group = algoritmo, color = algoritmo),
    inherit.aes = FALSE,
    linewidth = 0.9
  ) +

  geom_point(
    data = hist_all |> group_by(algoritmo) |> slice(1),
    aes(x = x1, y = x2, color = algoritmo),
    inherit.aes = FALSE,
    shape = 4,
    size = 3,
    stroke = 1.2
  ) +

  geom_point(
    data = hist_all |> group_by(algoritmo) |> slice(n()),
    aes(x = x1, y = x2, color = algoritmo),
    inherit.aes = FALSE,
    shape = 16,
    size = 3
  ) +
  coord_equal() +
  labs(
    x = "x1",
    y = "x2",
    fill = "f(x1,x2)",
    color = "Algoritmo",
    title = "Curvas de nível com trajetórias de diferentes métodos de gradiente"
  ) +
  theme_minimal()
```




Adam > RMSProp > Gradiente Normal > Gradiente com momento